

Unit 5

Planning – planning problem – Simple planning agent – Blocks world problem – Mean Ends analysis Learning - Machine learning - Learning concepts, methods and models Introduction to expert system – architecture of expert systems.

Planning

- The task of coming up with a sequence of actions that will achieve a goal is called planning.
- Planning Environments
 1. Classical Planning Environments
 - Fully observable, deterministic, finite, static and discrete.
 2. Non classical Planning Environments
 - Partially observable, stochastic with different algorithms and agent designs.

Planning Problem

- The planning problem is actually the question how to go to next state or the goal state from the current state. It involves two things 'how' and 'when'.
- The planning problem is defined with:

1. Domain model

- The domain model defines the actions along with the objects. It is necessary to specify the operators too that actually describe the action. Along with this, information about actions and state constraints while acting should also be given. This entirely formulates the domain model.

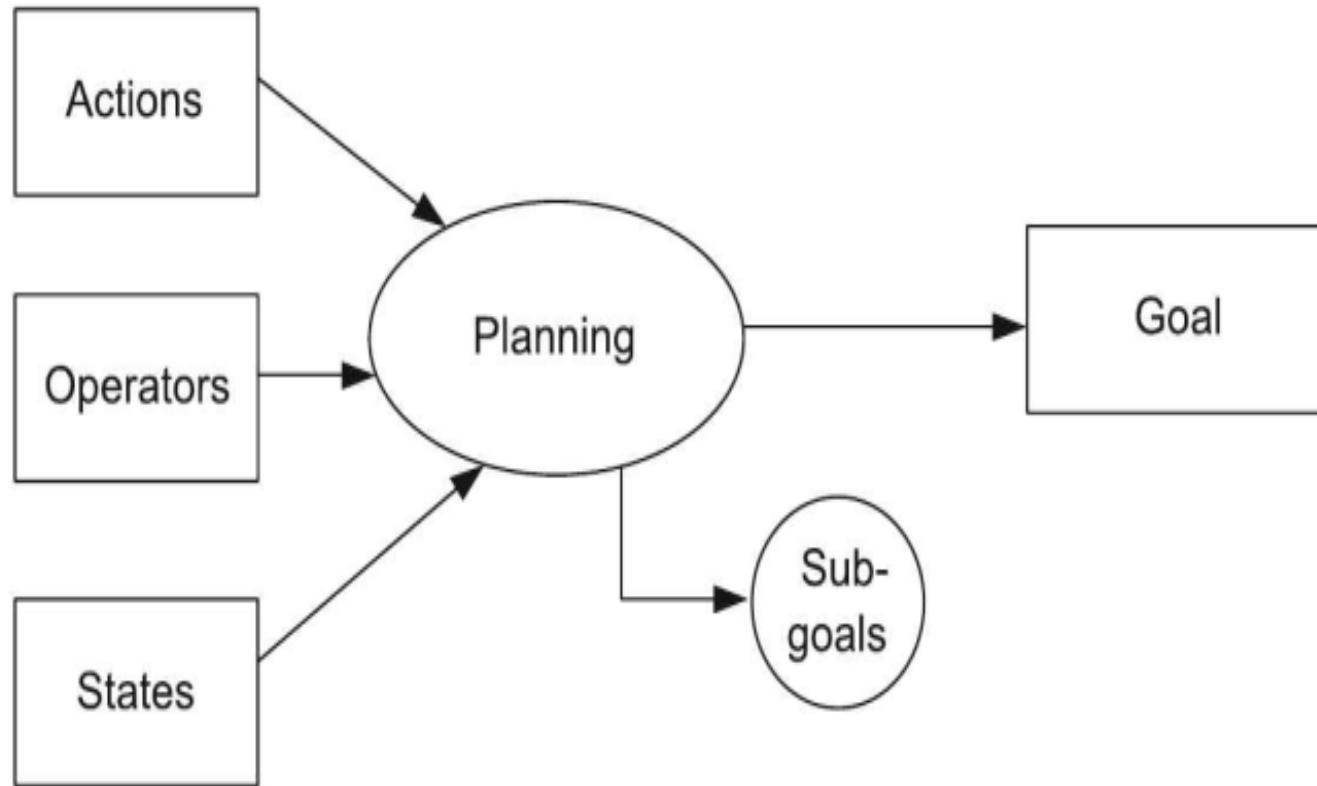
2. Initial state

- The initial state is the state where any action is yet to take place (the stage when the exam schedule is put up!).

3. Goal state (next state)

- The final state or the goal state is the state which the plan is intended to achieve.

Basic Planning Representation



- Planning representation involves
 - 1) State Representation
 - 2) Goal Representation
 - 3) Action Representation

State Representation

- A state is represented with a conjunction of positive literals using
 - Logical Propositions: $\text{Poor} \wedge \text{Unknown}$
 - FOL literals: $\text{At}(\text{Plane1}, \text{OMA}) \wedge \text{At}(\text{Plane2}, \text{JFK})$
- FOL literals must be ground & function-free
 - Not allowed: $\text{At}(x, y)$ or $\text{At}(\text{Father}(\text{Fred}), \text{Sydney})$
- Closed World Assumption
 - What is not stated are assumed false

Goal Representation

- Goal is a partially specified state
- A proposition satisfies a goal if it contains all the atoms of the goal and possibly others.
- Example: $\text{Rich} \wedge \text{Famous} \wedge \text{Miserable}$ satisfies the goal $\text{Rich} \wedge \text{Famous}$

Action Representation

- Action Schema

- Action name
- Preconditions
- Effects

$At(WHI, LNK), Plane(WHI),$
 $Airport(LNK), Airport(OHA)$

$Fly(WHI, LNK, OHA)$

$At(WHI, OHA), \neg At(WHI, LNK)$

- Example

Action($Fly(p, from, to)$,

PRECOND: $At(p, from) \wedge Plane(p) \wedge Airport(from) \wedge Airport(to)$

EFFECT: $\neg At(p, from) \wedge At(p, to)$)

- Sometimes, Effects are split into ADD list and DELETE list

Simple planning agent

- A simple planning agent is very similar to problem-solving agents in that it constructs plans that achieve its goals, and then executes them.
- The limitations of the problem-solving approach motivates the design of planning systems.
- To solve a planning problem using a state-space search approach we would let the:
 - initial state = initial situation
 - goal-test predicate = goal state description
 - successor function computed from the set of operators
 - once a goal is found, solution plan is the sequence of operators in the path from the start node to the goal node

Algorithm of a simple planning agent

1. Generate a goal to achieve
 2. Construct a plan to achieve goal from current state
 3. Execute plan until finished
 4. Begin again with new goal
- The agent first generates a goal to achieve, and then constructs a plan to achieve it from the current state.
 - Once it has a plan, it keeps executing it until the plan is finished, then begins again with a new goal.

Languages for Planning Problems

- STRIPS
 - Stanford Research Institute Problem Solver
 - Historically important
- ADL
 - Action Description Languages
- PDDL
 - Planning Domain Definition Language

Stanford Research Institute Problem Solver (STRIPS)

- Makes use of the first order predicates.
- STRIPS allows function-free literals.
- Example of a robot: The example involves a robot, a cup tea, guest and two rooms. We want the robot to get the tea and give it to the guest.

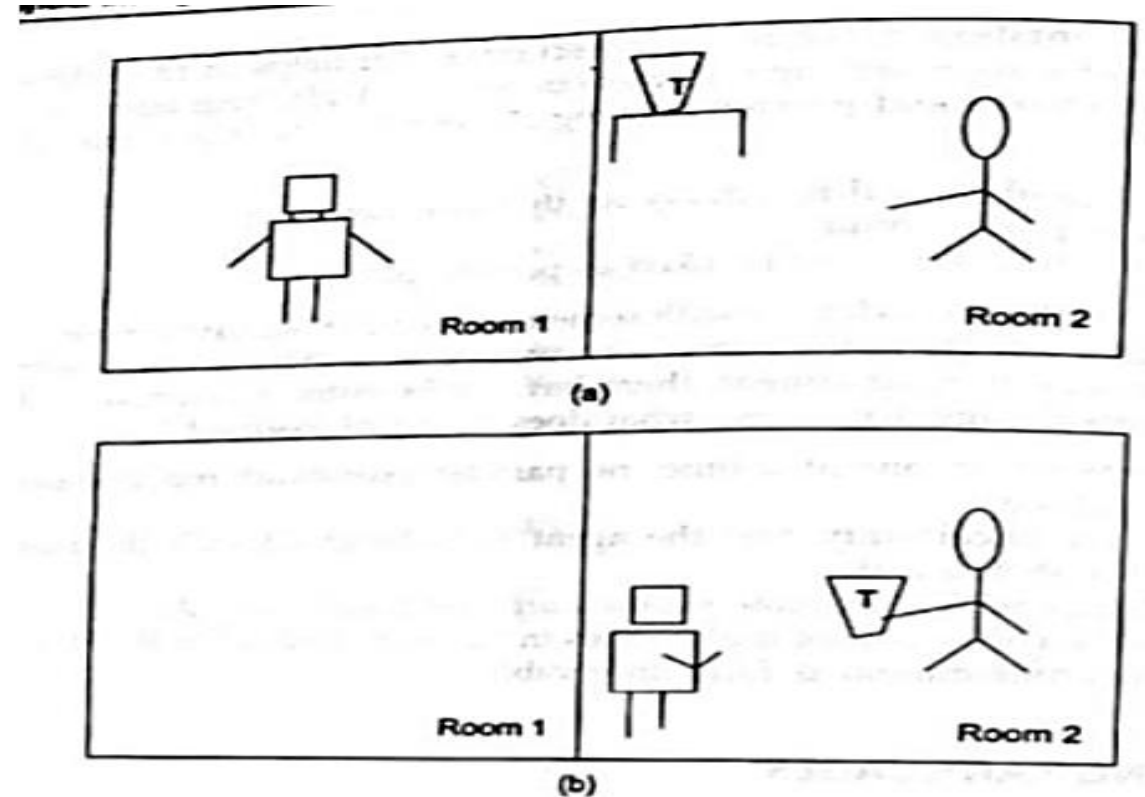


Figure 9.2 (a) Initial stage, (b) Goal state.

Action Description Language (ADL)

- STRIPS language lacks the expressive power.
- It can be extended very well to overcome some of the limitations and ADL does that.
- The properties of ADL are as follows:
 1. It allows negative literals.
 2. It makes use of quantified variables along with the disjunctions and the conjunctions.
 3. Conditional post-conditions are allowed.
 4. Variables with different types at the same time are allowed and also equality property is available.

Planning Domain ~~Definition~~ Language (PDDL)

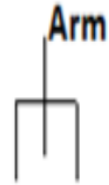
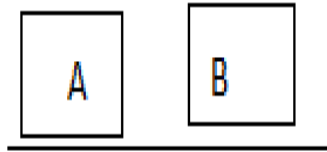
- It is a standardization of the planning languages.
- It is defined by the researchers as a standard language.
- We can say it is a superset of STRIPS and ADL that allows features like
 1. Objects can have type specifications.
 2. It can have negative pre-conditions.
 3. The add and delete lists can be conditional.
 4. In some cases, it also allows numeric values.

Block World

- There are 'N' number of Blocks resting on table with specified sequence.
- Goal is to arrange in desired sequence.
- Available moves
 - Put block on table
 - Put a block on another block top
- State is represented using sequence of blocks in current pos.

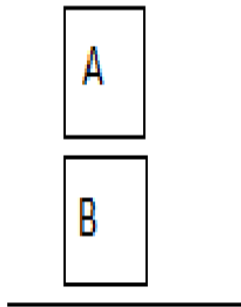
Block World

- Start State:



- Start State:
 - On(A, table)
 - On(B, table)
- Goal State:
 - On(A,B)

- Goal State:



Solution:

On(A,B)



Stack(A,B)

Preconditions:



~~Holding(A)~~
✓ Clear(B)

Pickup(A)

Preconditions:

Arm Empty ✓

On(A, Table) ✓

Clear(A) ✓

Means - Ends Analysis

- Means-Ends Analysis is problem-solving techniques used in Artificial intelligence for limiting search in AI programs.
- It is a mixture of Backward and forward search technique.
- The means -ends analysis process centers around finding the difference between current state and goal state.
- Search strategies are either reason forward or backward.

Steps

1. First, evaluate the difference between Initial State and final State.
2. Select the various operators which can be applied for each difference.
3. Apply the operator at each difference, which reduces the difference between the current state and goal state.

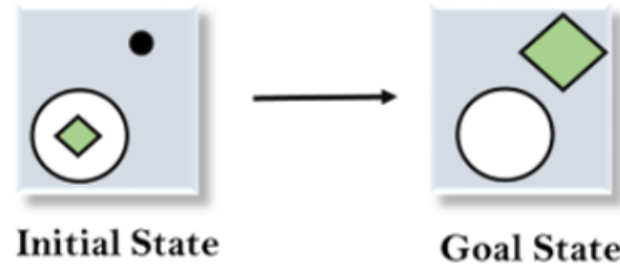
MEA Algorithm

Step 1: Compare CURRENT to GOAL, if there are no differences between both then return Success and Exit.

- Step 2: Else, select the most significant difference and reduce it by doing the following steps until the success or failure occurs.
- A) Select a new operator O which is applicable for the current difference, and if there is no such operator, then signal failure.
- B) Attempt to apply operator O to CURRENT. Make a description of two states.
 - i) O-Start, a state in which O's preconditions are satisfied.
 - ii) O-Result, the state that would result if O were applied in O-start.
- C) If
 - (First-Part <----- MEA (CURRENT, O-START)
 - And
 - (LAST-Part <----- MEA (O-Result, GOAL), are successful, then signal
 - Success and return the result of combining FIRST-PART, O, and LASTPART.

Example of Mean-Ends Analysis:

- Apply MEA to get the goal state.



- **Solution:**
 - To solve the above problem, first find the differences between initial states and goal states, and for each difference, generate a new state and will apply the operators. The operators we have for this problem are:
 - **Move**
 - **Delete**
 - **Expand**

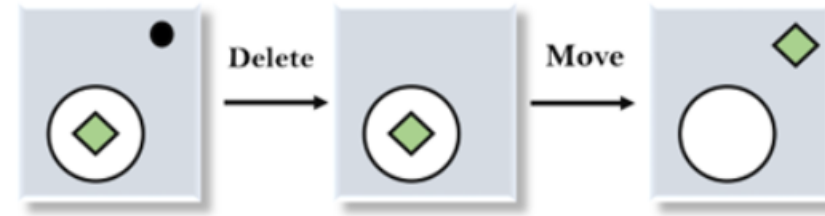
Example of Mean-Ends Analysis:

Step 1: Evaluate Initial State



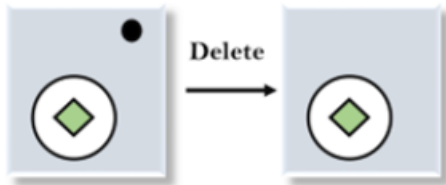
Initial state

Step 3: Apply Move Operator



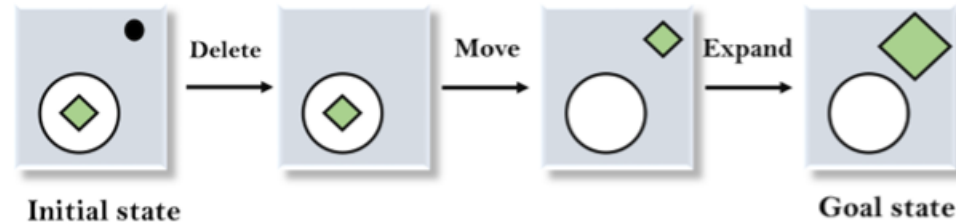
Initial state

Step 2: Apply Delete Operator



Initial state

Step 1: Apply Expand Operator



Initial state

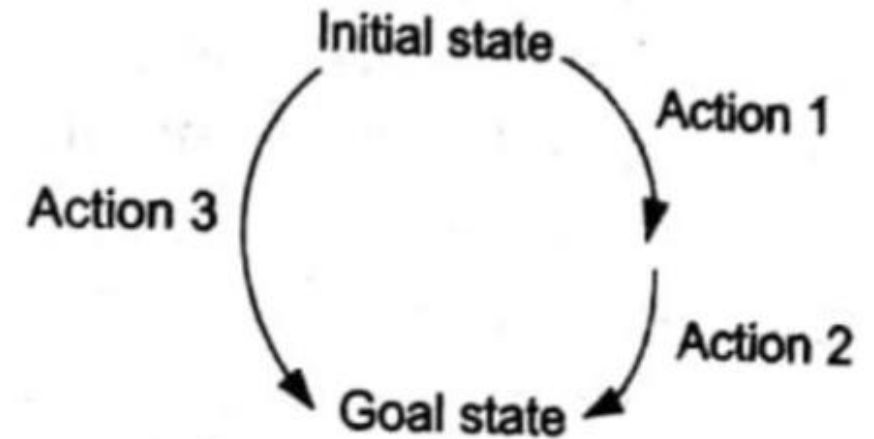
Goal state

Non-linear planning

- A plan that consists of sub-problems, which are solved simultaneously is said non-linear plan.
- In case of the goal stack planning there are some problems. To achieve any goal, it could have an impact on the one that has been achieved.

There is a concept of constraint posting that comes with non-linear planning. The constraint posting states that the plan can be built by

- 1. Addition of operators or suggesting operators
- 2. Ordering them
- 3. Binding the variables to the operators



Non-linear planning.

Conditional Planning

- Conditional planning has to work regardless of the outcome of an action.
- The outcome of actions cannot be determined so the environment is said to be nondeterministic.
- It's a way to deal with uncertainty by checking what is actually happening in the environment at predetermined points in the plan. (Conditional Steps)

Example:

- Check whether SFO airport (San Francisco International Airport) is open

Three kind of Environments

- Fully Observable - The agent always knows the current state
- Partially Observable - The agent knows only a certain amount about the actual state. (much more common in real world)
- Unknown - The agent knows nothing about the current state.

Conditional Planning in Fully Observable Environments

- Agent used conditional steps to check the state of the environment to decide what to do next.
- Plan information stores in a library

E Action(Left) → Clean v Right

- Syntax: If then plan_A else plan_B

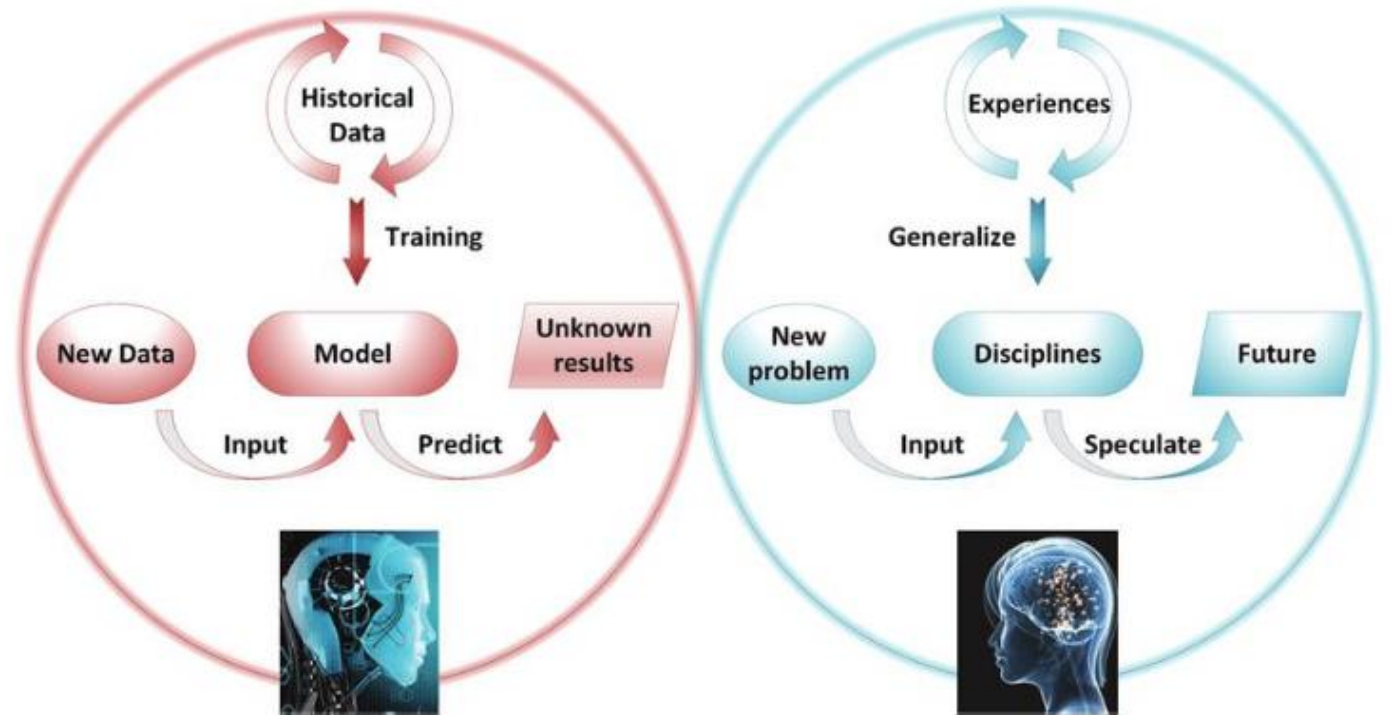
Reactive Planning

- Reactive planning is planning under uncertainty.
- Makes use of the if-then rules.
- The reactive planners are based on the concept that they should be able to handle an unknown situation too. So, the reaction rules are used that help them in doing so.
- A rule selection is based on the priority and a holding condition that maximises the priority.
- The rule which is at present in execution is said to be active whereas the ones with holding priority (we can call them possible competitors) are pre-active others are inactive.
- A B-tree structure is used in reactive planning, where the things are algorithm selects the rule. Sometimes, no rule can be selected. In such a case, dependent on the algorithm implementation for rule selection.

Machine Learning

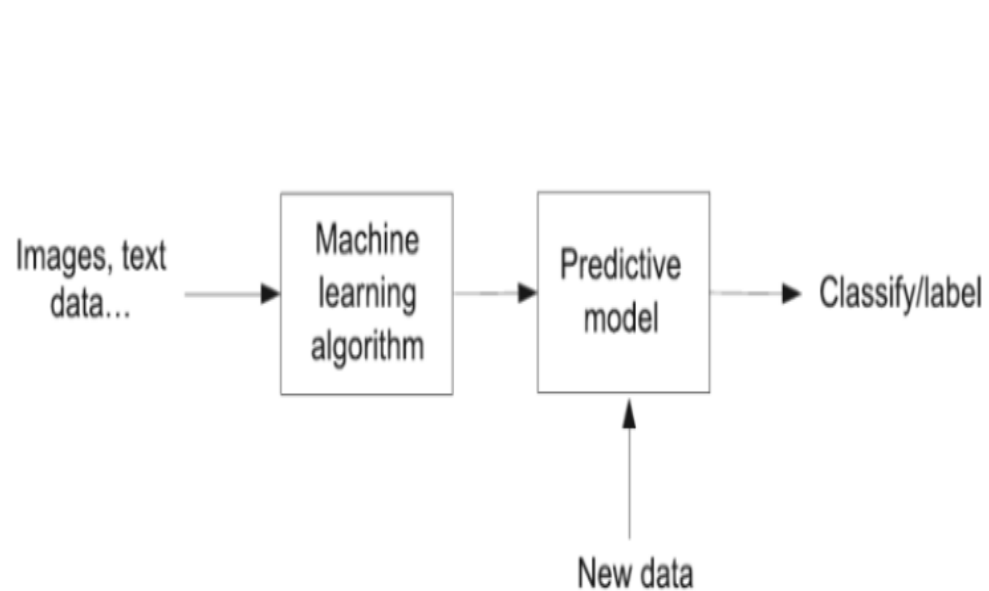
- Machine learning is building and exploring of methods for programming computer to make them Learn.

Machine learning Vs Human Learning



Machine Learning

- Machine Learning Approach



- Scope of ML



Automatic Translation



Traffic Prediction



Web Search and Recommendation Engines



Virtual Personal Assistants



Online Fraud Detection



Medical Diagnosis



Text & Speech Recognition



Email spam filtering



Image Recognition

Machine Learning

Goals of machine learning

- To produce learning algorithms with practical value.
- Development and enhancement of computer algorithms and models to meet the decision making requirements in practical scenarios.
- To facilitate in building Intelligent System (IS) that can be used in solving real time problems.

Challenges of Machine Learning

- Availability of limited learning data and unknown perspectives.
- Acquiring Accurate , compact and precise knowledge building.
- Require large working memory to store data.
- Focusing Too Much on Algorithms and Theories
- Monitoring and maintenance

Machine Learning

Computational structure used in Machine learning:

1. Functions.
2. Logic programs and rule sets
3. Finite state machines
4. Grammars
5. Problem solving system

Types of Machine Learning

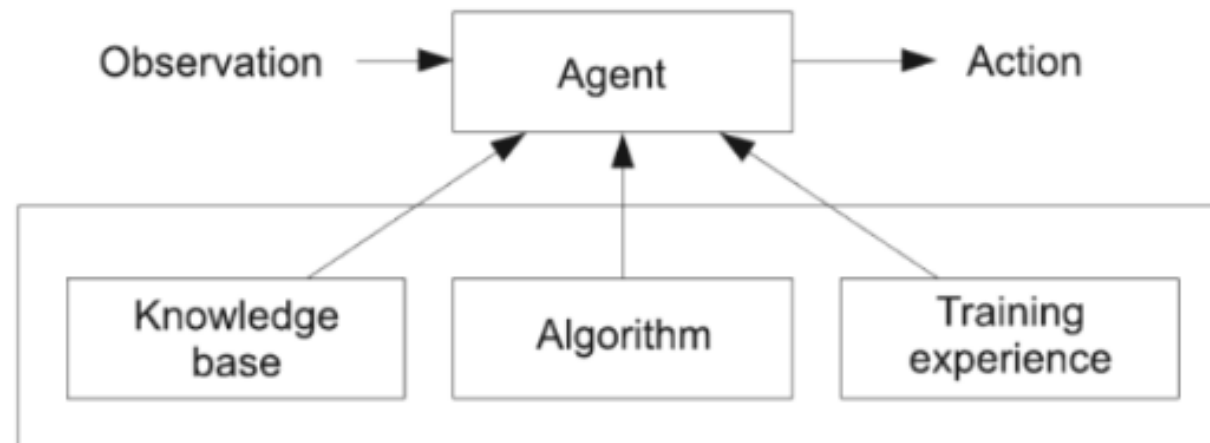
- Rote learning
- Learning from observations
 - Learning from agents
 - Inductive learning
 - Decision tree learning
- Supervised learning
- Unsupervised learning
- Semi-supervised learning
- Ensemble learning
- Discovery based learning
- Learning by problem solving

ML - Rote Learning

- Rote learning is a rudimentary form of learning, which focusses on memorization.
- Here, the values are stored so that these are not re-computed.
- When computations are more expensive than re-computing from scratch, this strategy can save a significant amount of time.
- Selective paging and caching are some of the examples of rote learning.

ML - Learning from agents

- An agent is defined the computational entity which is capable of receiving the environment and can act based on the situation.
- The agent is composed of learning element performance element and a curiosity element.
- Based on the coordination between these elements the outcome of the agent's behavior is measured.

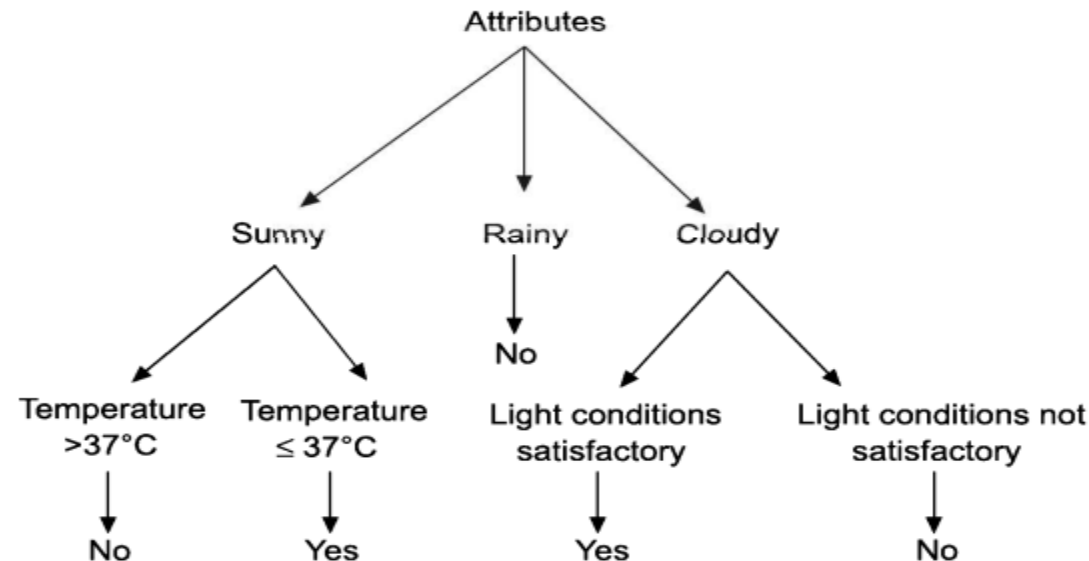


ML - Inductive Learning

- Inductive learning involves learning generalized rules from specific examples (can think of this as the “inverse” of deduction)
- Main task: given a set of examples, each classified as positive or negative produce a concept description that matches exactly the positive examples.
- The examples are coded in some representation language, e.g. they are coded by a finite set of real-valued features.
- The concept description is in a certain language that is presumably a superset of the language of possible example encodings.
- A “correct” concept description is one that classifies correctly ALL possible examples, not just those given in the training set.

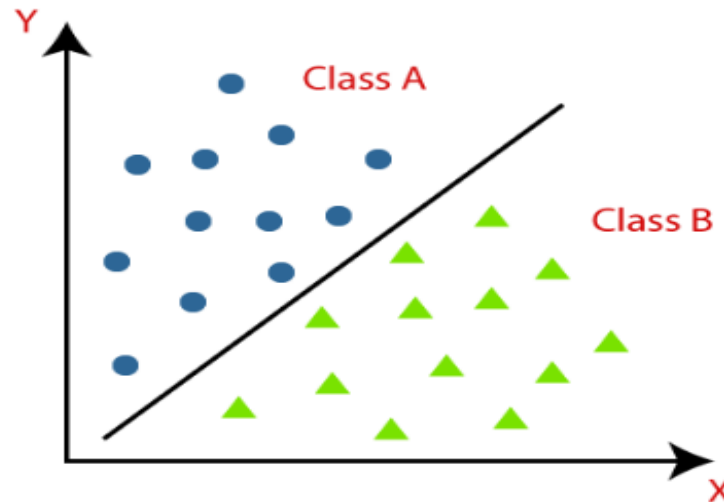
ML - Decision Tree Learning

- The learned function is represented by a decision tree.
- In terms of programming it is also represented as if then rules.
- Decision tree depicts the simple learning from the observation, method.
- Based on the observation, at every node, decision is taken.



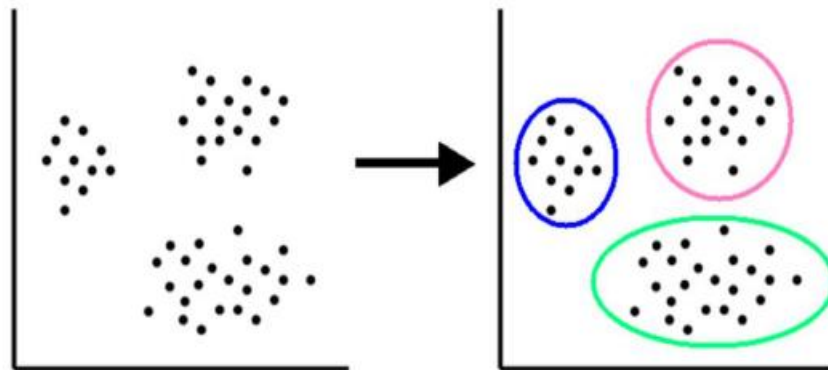
ML - Supervised learning

- Uses labelled dataset.
- Supervised learning is the learning algorithm that is provided with the set of training data and the algorithm further induces the classifier to classify the unseen or new data.
- A line (hyperplane) which is generated after learning separating two classes class A and class the in two parts the classifier and the decision-making engine minimize the false positives and false negative.



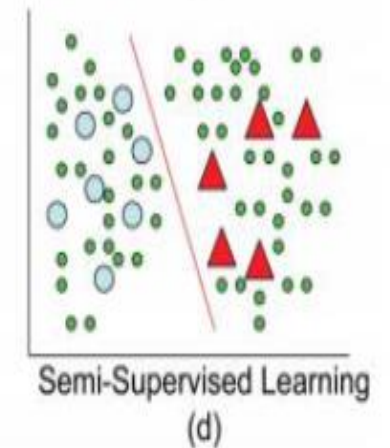
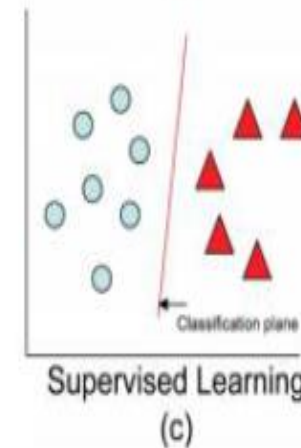
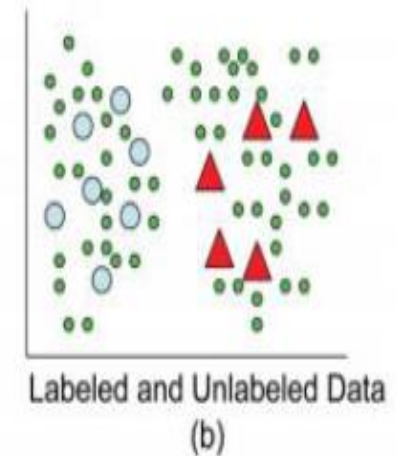
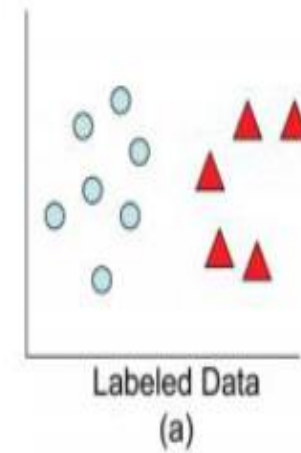
ML - Unsupervised Learning

- Use unlabeled dataset.
- Learning is more based on similarities and differences which are visible. These differences and similarities are mathematically represented in unsupervised learning
- Grouping and categorization of the objects is based on the understanding of similarities and visualization of their relations
- Unsupervised learning performs hierarchical clustering.

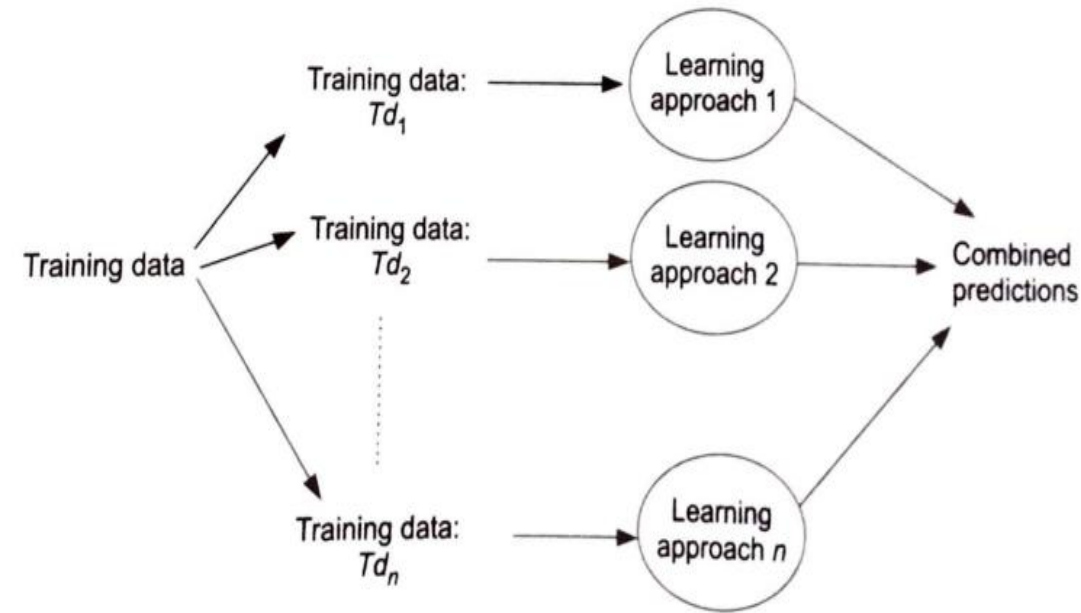


ML - Semi supervised Learning

- Semi supervised learning is developed to cope up with the issues of learning in supervised or unsupervised mode in isolation.
- Semi-supervised learning tries to learn from the labelled as well as unlabeled data.
- Let U be a set of unlabeled data and L be a set of labelled data.
- As the learning process the learning approach identifies the unlabeled data U with reference to a labelled data L and keeps on labelling the unlabeled data.
- This method is also called as self training in semi supervised learning.



ML - Ensemble Learning



Ensemble method: The concept.

- Ensemble learning method is the one where multiple learners or learning algorithms are trained.
 - In ensemble learning method the whole collection or ensemble of hypothesis is selected from the hypothesis space and their predictions are combined.
 - In this approach, the learners or referred to as base learners.
 - The most commonly used ensemble learning methods are
- 1. Boosting:**
- Boosting can probably be defined as the method for generating accurate predictions by combining the rules that are comparatively inaccurate.
 - Boosting works on the weighted training sets. The weights of the training example reflects the importance of training examples.
- 2. Bagging:**
- In Bagging, the training data is resampled. This is referred to as bootstrap sampling, where the training data with replacement is taken in the learning approaches.

ML

Discovery-based Learning

- In case of discovery learning the learning, takes place without any help of teacher or supervisor.
- It is called unsupervised.
- It can be said to be inquiry-based learning, where the learner on the basis of past experience and knowledge tries to discover the outcomes.

Learning by Problem Solving

- In learning from problem solving, various parameters related to solution and problem are considered.
- These parameters are used and effectively desirability of a particular outcome or decision is determined.
- It is typically based on the way a problem is solved and the outcomes at every step.
- Even in pattern-based decision- making, the correct category is determined based on the previous results.

ML - Learning Methods

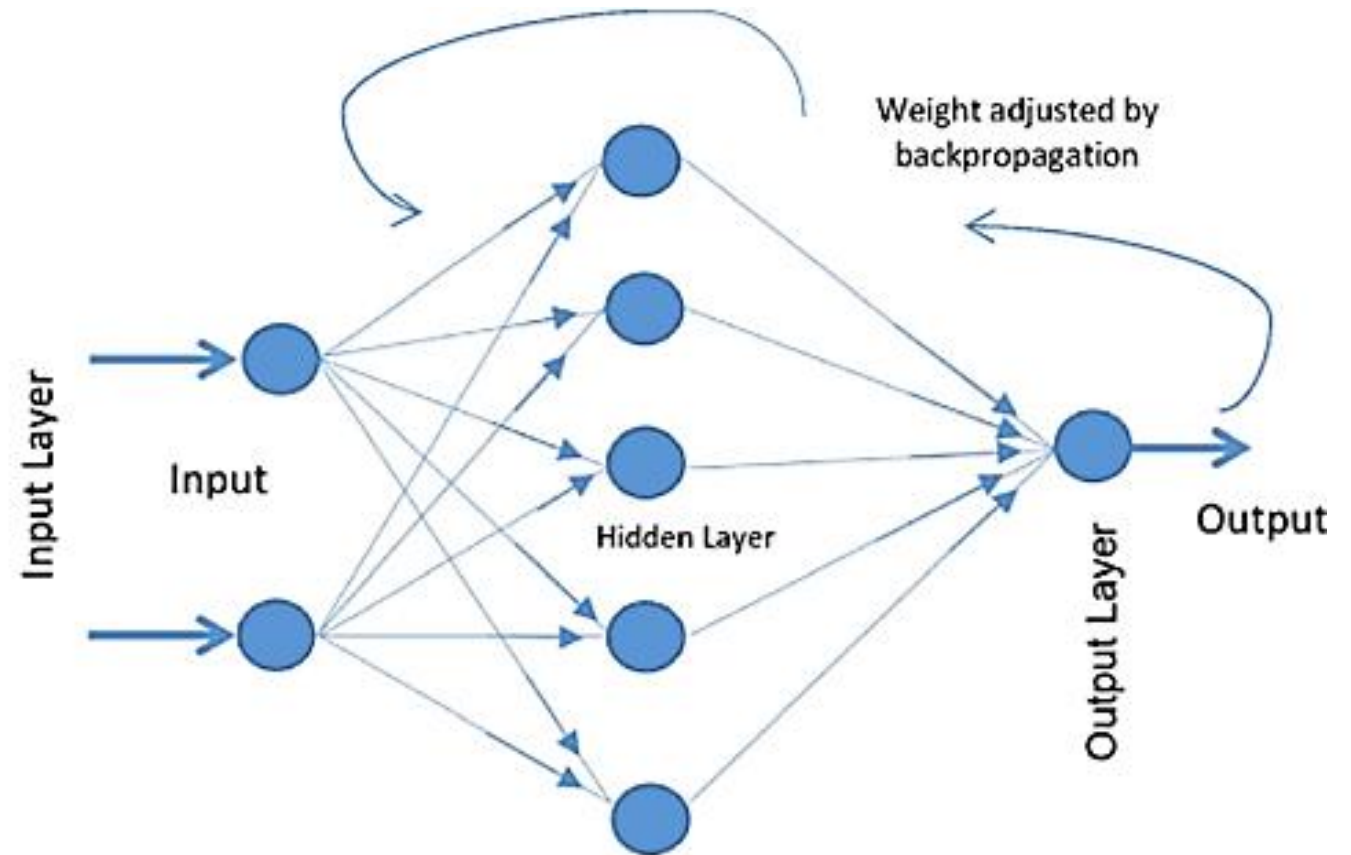
- Artificial neural network-based learning - Back propagation
- Support vector machines
- Reinforcement learning
- Adaptive learning
- Multi agent based learning
- Distributed learning
- Speedup learning

Artificial neural network based learning

- An Artificial Neural Network (ANN) is a computational model inspired by the human brain's neural structure.
- It consists of interconnected nodes (neurons) organized into layers.
- Information flows through these nodes, and the network adjusts the connection strengths (weights) during training to learn from data, enabling it to recognize patterns, make predictions, and solve various tasks in machine learning and artificial intelligence.

ANN - Back propagation Algorithm

- Back-propagation is the essence of neural net training.
- It is the method of fine-tuning the weights of a neural net based on the error rate obtained in the previous epoch (i.e., iteration).
- Proper tuning of the weights allows you to reduce error rates and to make the model reliable by increasing its generalization.

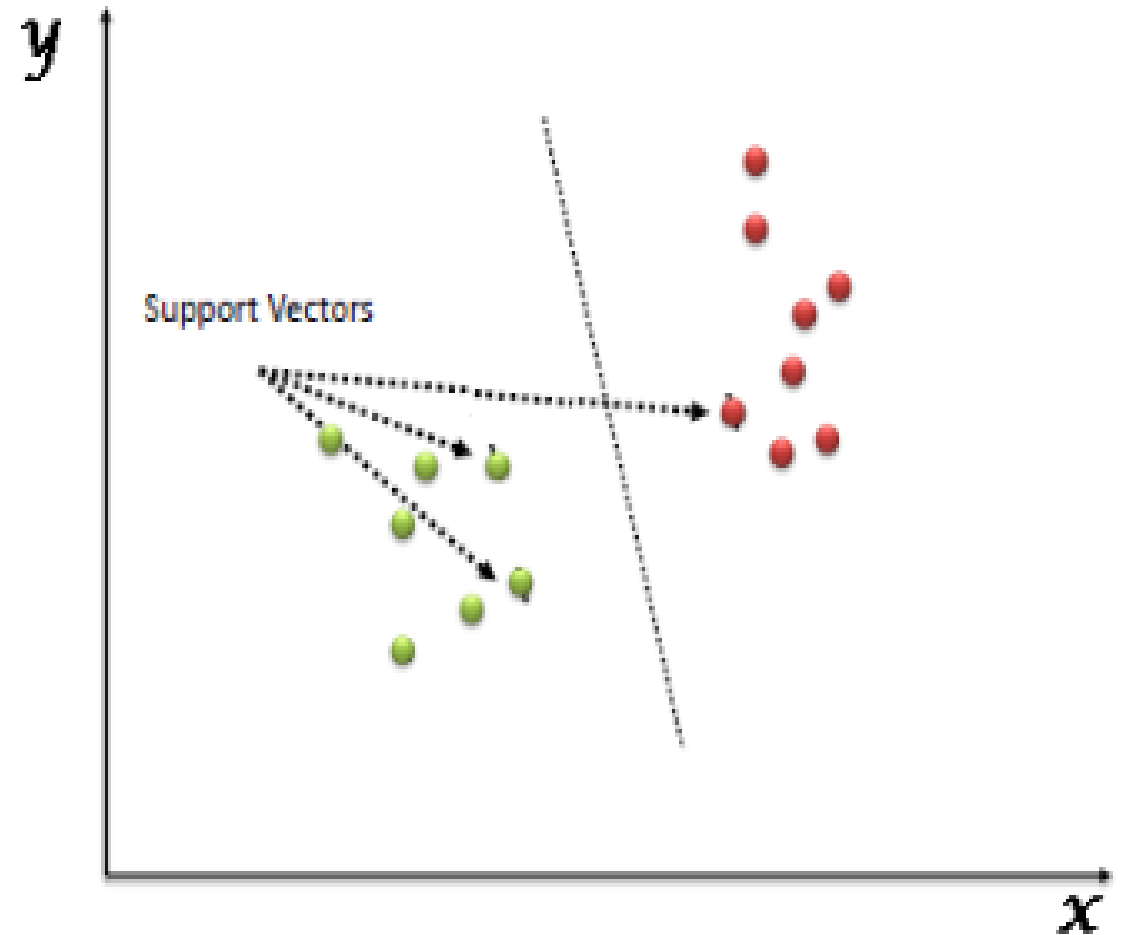


ANN - Back propagation Algorithm

1. Inputs X, arrive through the preconnected path
2. Input is modeled using real weights W. The weights are usually randomly selected.
3. Calculate the output for every neuron from the input layer, to the hidden layers, to the output layer.
4. Calculate the error in the outputs.
$$\text{ErrorB} = \text{Actual Output} - \text{Desired Output}$$
5. Travel back from the output layer to the hidden layer to adjust the weights such that the error is decreased.
6. Keep repeating the process until the desired output is achieved.

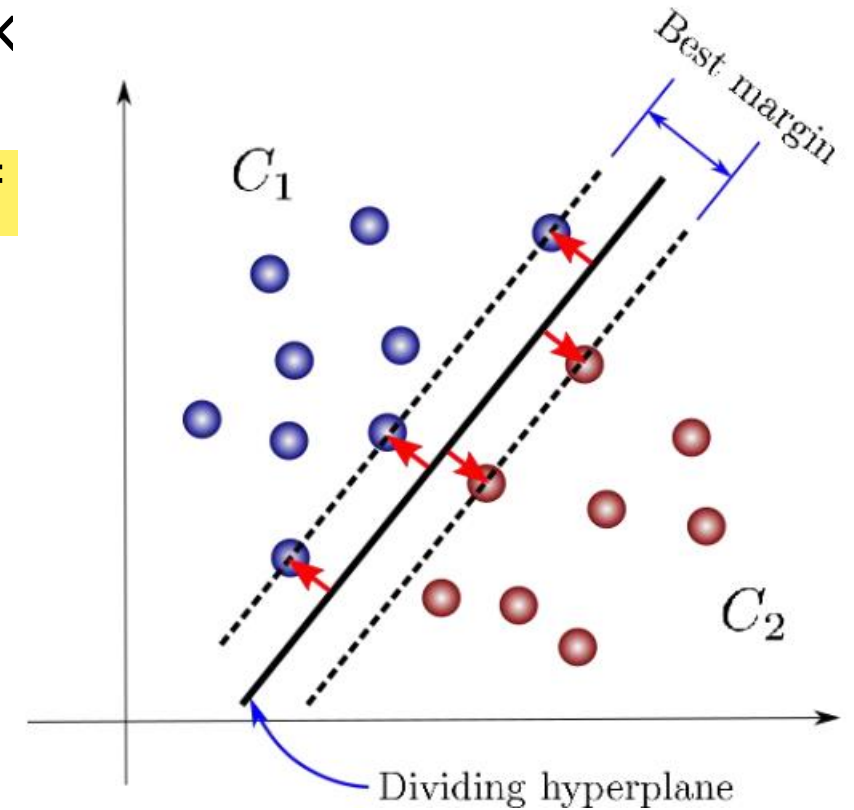
Support Vector Machine (SVM)

- Support vector machines (SVM) are supervised learning models with associated learning algorithms that analyze data used for classification and regression analysis.
- It is a machine learning approach.
- They analyze the large amount of data to identify patterns from them.
- SVMs are based on the idea of finding a hyperplane that best divides a dataset into two classes, as shown in the image below.



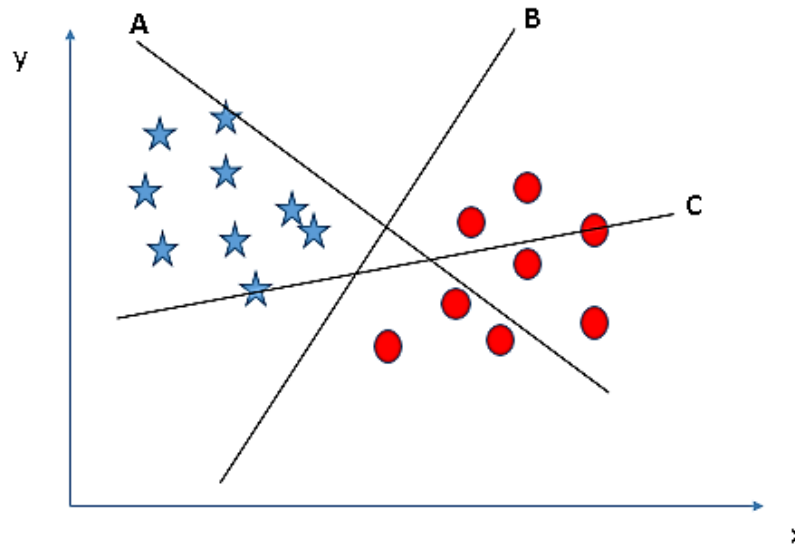
Hyperplane

- As a simple example, for a classification task with only two features, hyperplane is a line that linearly separates and classifies a set of data.
- Intuitively, the further from the hyperplane our data points lie, the more confident we are that they have been correctly classified.
- We therefore want our data points to be as far away from the hyperplane as possible, while still being on the correct side of it.
- So when new testing data are added, whatever side of the hyperplane it lands will decide the class that we assign to it.



Identify the right hyperplane (Scenario-1):

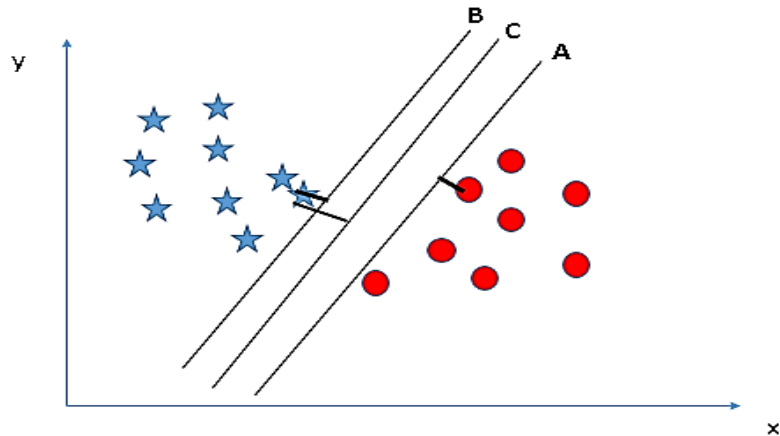
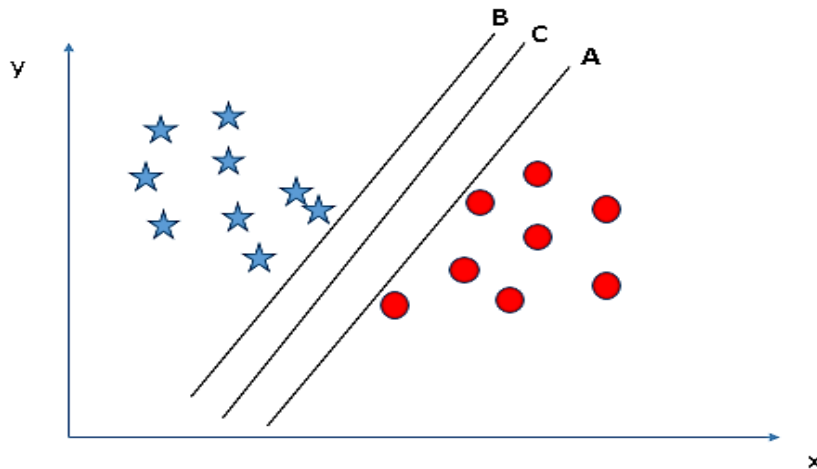
- Here, we have three hyperplanes (A, B and C). Now, identify the right hyperplane to classify star and circle.



Hyperplane “B” has excellently performed this job.

Identify the right hyperplane (Scenario-2):

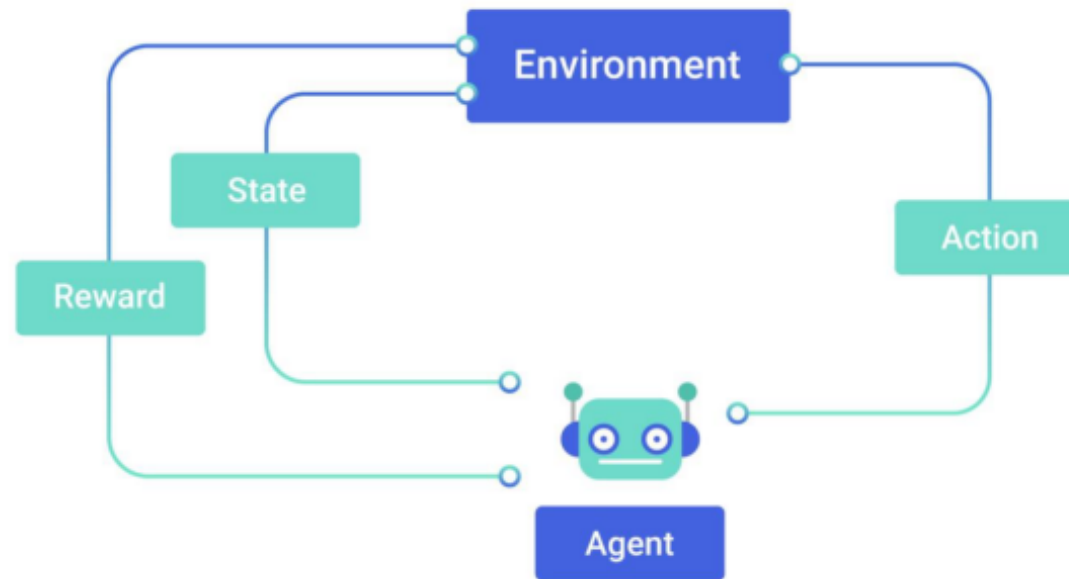
- Here, we have three hyperplanes (A, B and C) and all are segregating the classes well. Now, how can we identify the right



- Here, maximizing the distances between nearest data point (either class) and hyperplane will help us to decide the right hyperplane.
- This distance is called as Margin. Let's look at the second fig
- We can see that the margin for hyperplane C is high as compared to both A and B. Hence, we name the right hyperplane as C.
- Another lightning reason for selecting the hyperplane with higher margin is robustness. If we select a hyperplane having low margin then there is high chance of misclassification.

Reinforcement Learning

- Learning from interaction with an environment to achieve some long-term goal that is related to the state of the environment
- The goal is defined by reward signal, which must be maximised.
- Agent must be able to partially/fully sense the environment state and take actions to influence the environment state
- The state is typically described with a feature vector



Reinforcement Learning Systems

- Reinforcement learning systems have 4 main elements:

1. Policy-A policy is a mapping from the perceived states of the environment to actions to be taken when in those states

- A reinforcement learning agent uses a policy to select actions given the current environment state
- An on-policy agent learns only about the policy that it is executing
- An off-policy agent learns about a policy or policies different from the one that it is executing.

2. Reward signal

- The reward signal defines the goal
- On each time step, the environment sends a single number called the reward to the reinforcement learning agent,
- The agent's objective is to maximize the total reward that it receives over the long run
- The reward signal is used to alter the policy

Reinforcement Learning Systems

3. Value function

- The value of a state is the total amount of reward an agent can expect to accumulate over the future, starting in that state.
- Values must continually be re-estimated from the sequence of observations that an agent makes over its lifetime.

4. Optional model of the environment

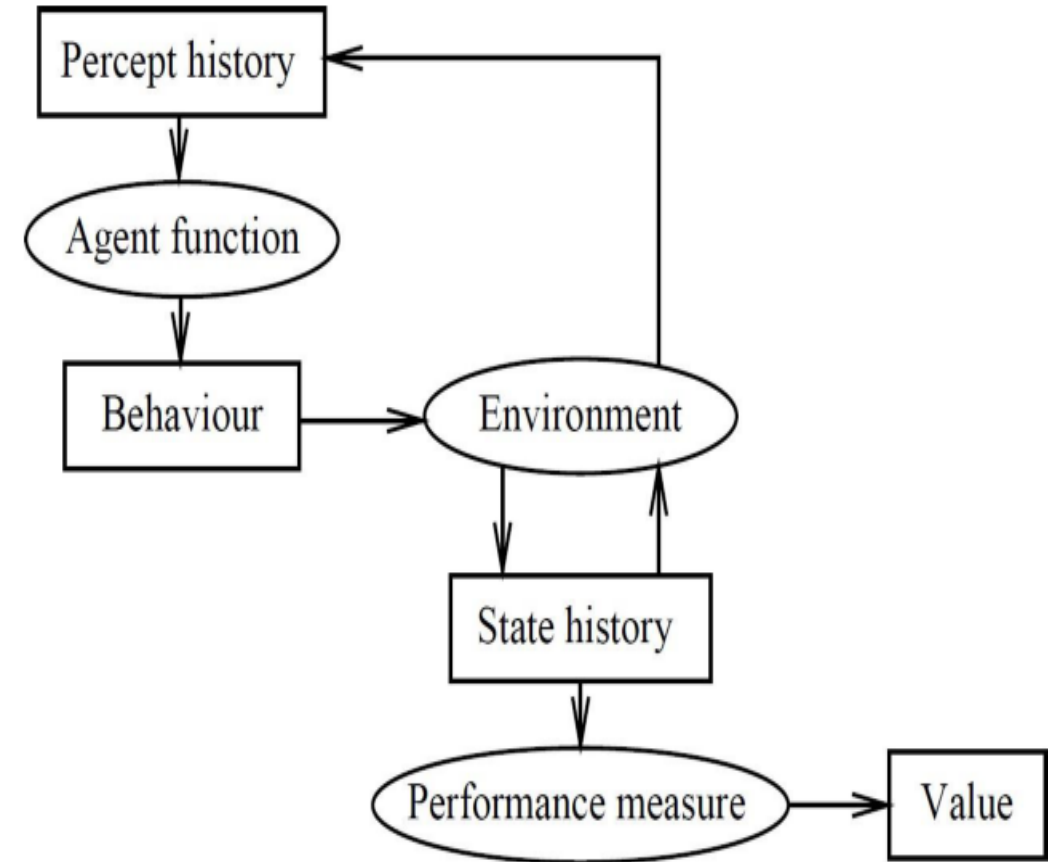
- A model of the environment allows inferences to be made about how the environment will behave.
- **Example:** Given a state and an action to be taken while in that state, the model could predict the next state and the next reward.
- Models are used for planning, which means deciding on a course of action by considering possible future situations before they are experienced

Adaptive learning

- Adaptive machine learning algorithms are the machine learning models, where the changes in the environment help in selecting the algorithm or learning method.
- The adaptive learning solves some of the complex problems for which a single learning method is not enough.
- This method is even more appropriate when the environment is continuously changing and real time response is expected.

Multi-Agent System

- A system with multiple autonomous entities, with distributed information, computational ability, and possibly divergent interests.
- A multiagent system is one that consists of a number of agents, which interact with one-another
- In the most general case, agents will be acting on behalf of users with different goals and motivations
- To successfully interact, they will require the ability to cooperate, coordinate, and negotiate with each other, much as people do

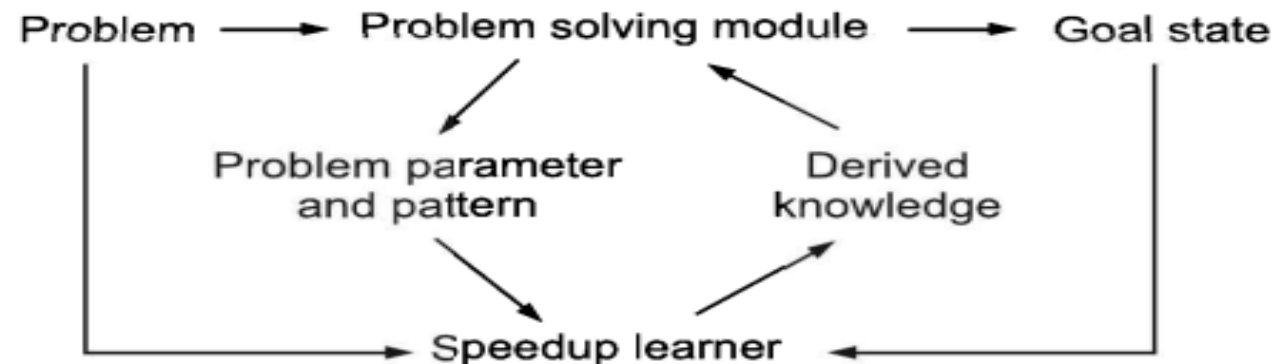


Distributed learning

- In distributed learning the task of learning is distributed.
- Need for distributed learning - Arises due to large data sets and time constraints.
- More than one agent in different parts of the data set.
- There will be distributed learning algorithms taken part in each partition to get the desired outcome, which would then be combined.
- Multi agent systems can be thought of as a subset of distributed learning.

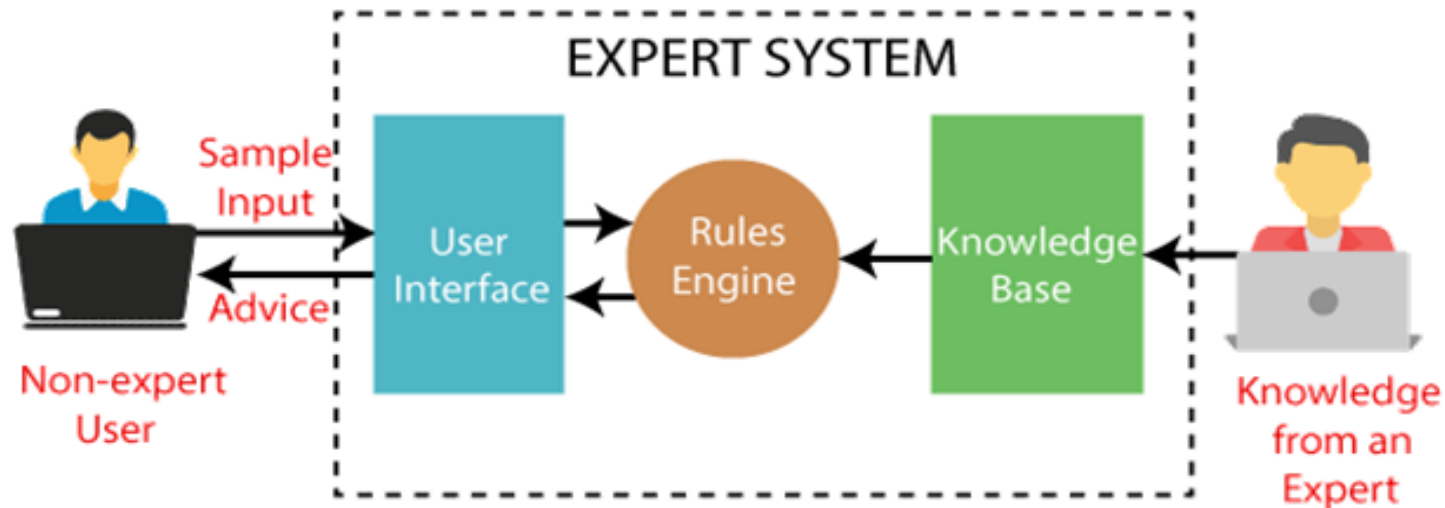
Speedup learning

- Speedup learning typically deals with speeding up problem solving by effective use of problem solving experience.
- Hence, prior problem solving experience is an input for speedup learning.
- In this learning,
 1. There is no interaction with the environment.
 2. New problems cannot be solved.
- So, speedup learning accelerates the process based on the previous experiences and prior observations.



Expert System

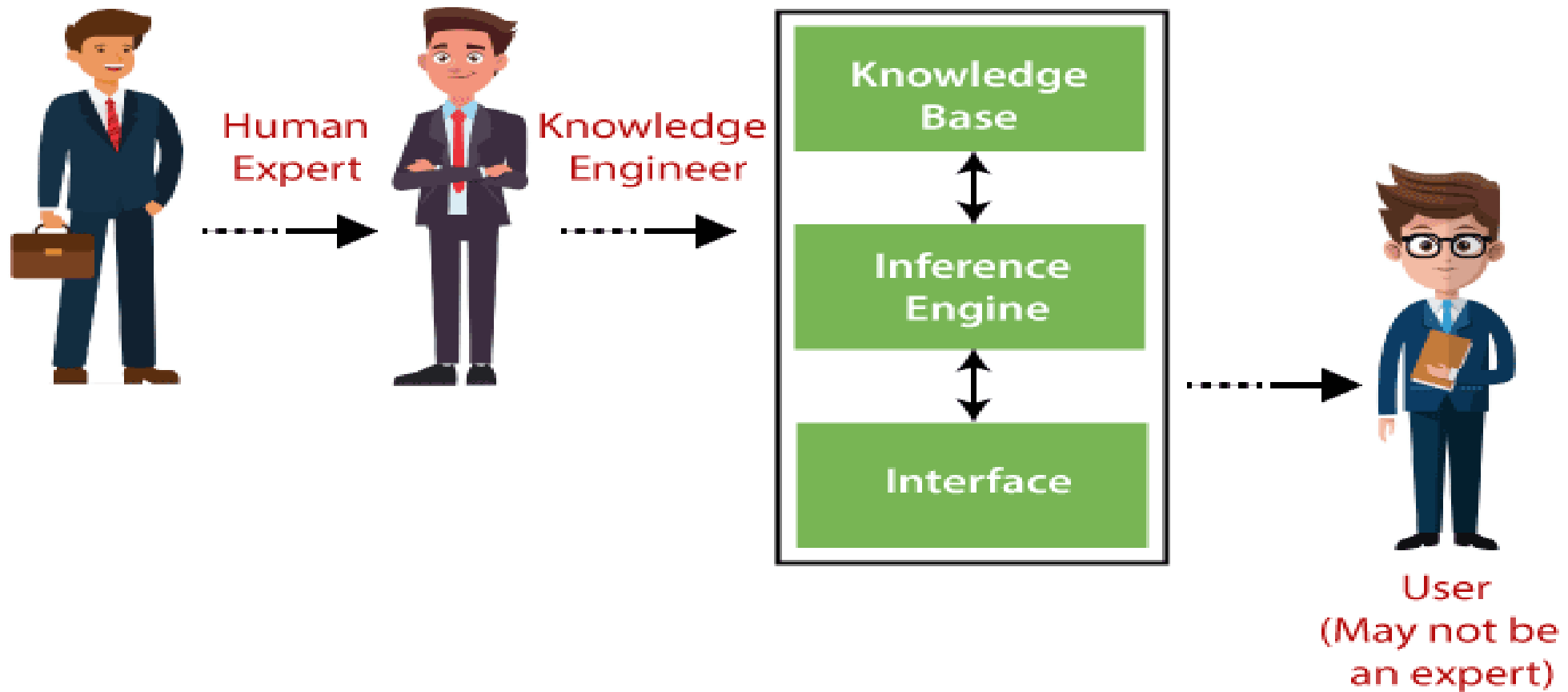
- An expert system is a computer program that is designed to solve complex problems and to provide decision-making ability like a human expert.
- It performs this by extracting knowledge from its knowledge base using the reasoning and inference rules according to the user queries.



Characteristics of Expert System

- **High Performance:** The expert system provides high performance for solving any type of complex problem of a specific domain with high efficiency and accuracy.
- **Understandable:** It responds in a way that can be easily understandable by the user. It can take input in human language and provides the output in the same way.
- **Reliable:** It is much reliable for generating an efficient and accurate output.
- **Highly responsive:** ES provides the result for any complex query within a very short period of time.

Components of Expert System



Components of Expert System

1. User Interface

- With the help of a user interface, the expert system interacts with the user, takes queries as an input in a readable format, and passes it to the inference engine.
- After getting the response from the inference engine, it displays the output to the user.
- In other words, it is an interface that helps a non-expert user to communicate with the expert system to find a solution.

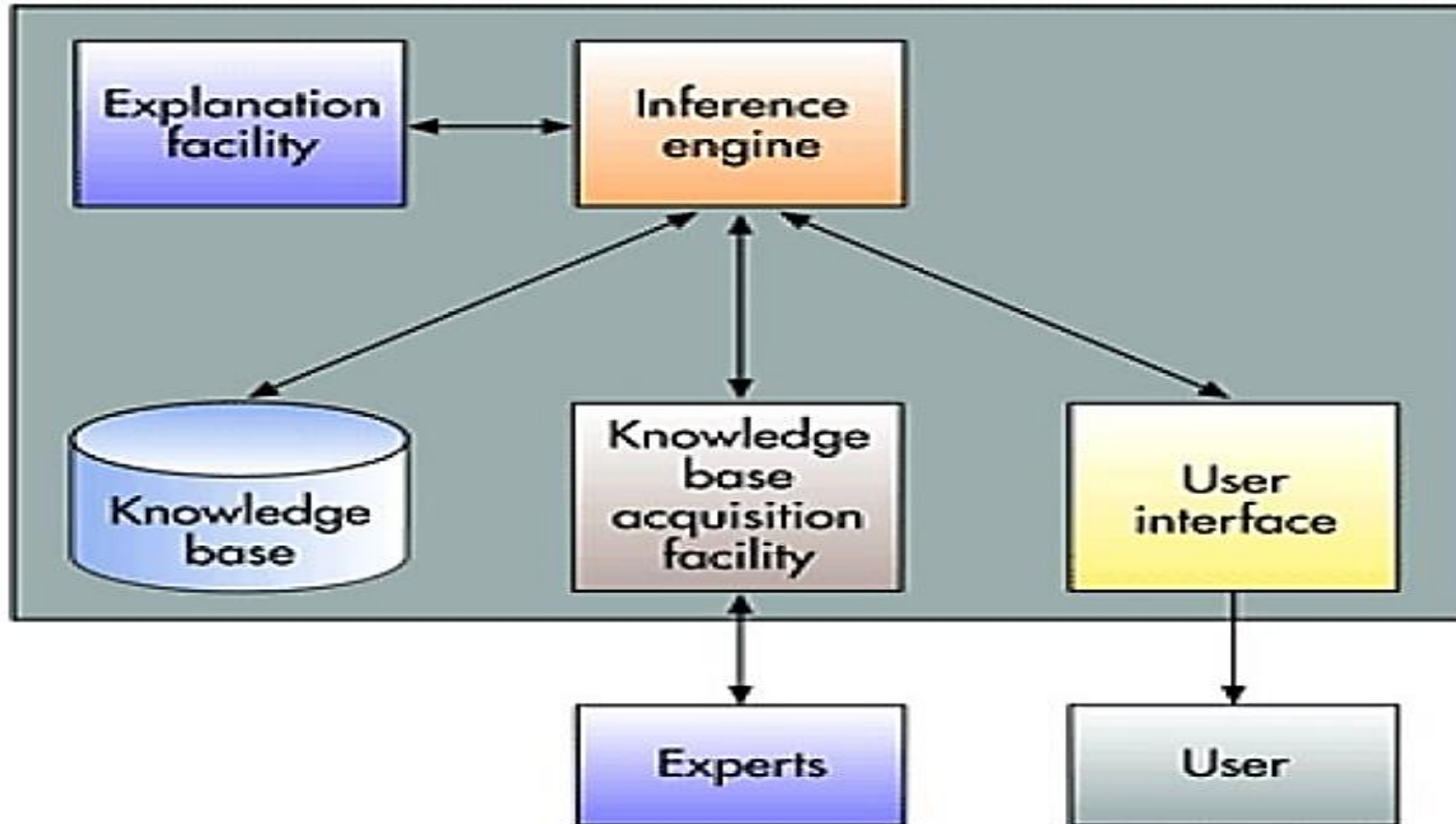
2. Inference Engine(Rules of Engine)

- The inference engine is known as the brain of the expert system as it is the main processing unit of the system.
- It applies inference rules to the knowledge base to derive a conclusion or deduce new information.
- It helps in deriving an error-free solution of queries asked by the user.

3. Knowledge Base

- The knowledgebase is a type of storage that stores knowledge acquired from the different experts of the particular domain.
- It is considered as big storage of knowledge.
- The more the knowledge base, the more precise will be the Expert System.

Architecture of Expert System



Architecture of Expert System

Explanation Facility

- Allows a user to understand how the expert system arrived at certain conclusions or results.
- For example: it allows a doctor to find out the logic or rationale of the diagnosis made by a medical expert system.

Inference Engine

- Seeks information and relationships from the knowledge base and provides answers, predictions, and suggestions the way a human expert would.

Knowledge Base

- Stores all relevant information, data, rules, cases, and relationships used by the expert system.
- It uses rules, If-then statements, fuzzy logic

Knowledge acquisition facility

- Provide convenient and efficient means of capturing and storing all the components of the knowledge base.
- Acts as an interface between experts and the knowledge base.

User Interface

- Specialized user interface software employed for designing, creating, updating, and using expert systems.
- The main purpose of the user interface is to make the development and use of an expert system easier for users and decision makers

Advantages and Limitations of Expert Systems

Advantages	Limitations
1. Permanent and not dependent on its use	1. Lacks creativity and commonsense
2. Rapid and easy transfer of knowledge between expert systems	2. Non-adaptable without human intervention
3. Easy to document the knowledge	3. Lacks sensory input of the problem
4. Relatively inexpensive	4. Narrow focus